

## Quantum Mechanics-1 (Assignment-1)

(Dated: September 2, 2024)

1. Consider a finite square well potential

$$V(x) = \begin{cases} -V_0 & \text{for } -a < x < a \\ 0 & \text{for } |x| > a \end{cases} \quad (1)$$

- (a) Show that, for large value of  $z_0 = \frac{a}{\hbar} \sqrt{2mV_0}$ , i.e., wide and deep well,  $E_n + V_0 \approx \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2}$ .  
(b) Show that there always exists at least one bound state for arbitrarily small value of  $z_0$ .

2. The potential for a finite spherical well is given by

$$V(r) = \begin{cases} -V_0 & \text{for } r \leq a \\ 0 & \text{for } r > a \end{cases} \quad (2)$$

- (a) Show that there exists no bound state if  $V_0 a^2 < \frac{\pi^2 \hbar^2}{8m}$ .  
(b) Derive the range of ground state energy.

3. The potential for a simple harmonic oscillator is given by  $V(x) = \frac{1}{2} m \omega^2 x^2$ . Taking  $y = \sqrt{\frac{m\omega}{\hbar}} x$ , we know from our class that  $\psi(y) \sim e^{-y^2/2}$  when  $y \rightarrow \infty$ . To get the wave-function for whole range of  $y$ , let us take  $\psi(y) = \phi(y) e^{-y^2/2}$ . Show that near  $y = 0$ , we get  $\phi(y) = H_n(y)$  where  $H_n(y)$  is Hermite polynomial.

4. Angular momentum operator in spherical polar coordinate can be written as  $L^2 = -\hbar^2 \left[ \frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin^2 \theta} \partial_\phi^2 \right]$ . To solve the eigenvalue equation  $L^2 \Psi = \hbar^2 \lambda \Psi$ , let us assume,  $\Psi(\theta, \phi) = \Theta(\theta) \Phi(\phi)$ .

- (a) Show that,  $\Phi(\phi) = e^{im\phi}$  where,  $m$  is an integer.  
(b) Show that,  $\Theta(\theta)$  satisfies

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left( \sin \theta \frac{d}{d\theta} \Theta \right) + \left( \lambda - \frac{m^2}{\sin^2 \theta} \right) \Theta = 0 \quad (3)$$

- (c) Perform a change of variable  $y = \cos \theta$  which transforms  $\Theta(\theta) \rightarrow \tilde{\Theta}(y)$ . Derive the differential equation satisfied by  $\tilde{\Theta}(y)$ .

- Solve this differential equation for  $\tilde{\Theta}(y)$  near  $y = \pm 1$ .
- Show that,  $\lambda = l(l+1)$  where  $l$  is some non-negative integer.
- Prove that, for  $m = 0$ ,  $\tilde{\Theta}(y) = P_l(y)$  where  $P_l(y)$  is the Legendre Polynomial.
- By performing  $m$ th order derivative on  $P_l(y)$  with respect to  $y$ , show that  $\tilde{\Theta}(y) = P_l^m(y)$  where  $P_l^m(y) = (1-y^2)^{|m|/2} \left( \frac{d}{dy} \right)^{|m|} P_l(y)$  is the Associated Legendre Polynomial.